

TECHNICAL REPORT

Nayi Disha · WasteLoop

An AI + IoT municipal waste management platform — working prototype architecture, database design, API surface, and end-to-end verified demo.

Prototype implementation of the architecture described in `dustbin_arch.md`

Backend: TypeScript · Fastify · Prisma · PostgreSQL | Frontend: React · Vite · Tailwind

Routing: self-hosted OSRM + nearest-neighbor/2-opt heuristic | Demo region: Central New Delhi

Generated 2026-08-19

1. Executive Summary

This report documents a working, runnable prototype of Nayi Disha (WasteLoop): a TypeScript/Fastify backend, a PostgreSQL database (via Prisma), a minimal React dashboard, and a real routing engine (OSRM), implementing the full citizen-facing loop — *scan* → *AI classify* → *time-limited token* → *triple-lock verification at a bin* → *reward release* — plus the municipal side: simulated IoT sensor telemetry, an admin command center, and dynamic multi-stop route optimization for collection vehicles.

14/14 backend smoke tests passing

0 npm audit vulnerabilities

Real OSRM road-network routing

Full golden path verified in-browser

Scope of this prototype

✓ Fully implemented

- Accounts, JWT + bcrypt auth, role-based access (citizen / admin)
- Full scan → token → triple-lock → throw → reward chain, including anti-gaming failure paths (expired token, GPS mismatch, missing evidence)
- Sensor telemetry ingestion, history storage, and a device-auth contract that a real device can use unmodified
- Real road-network route optimization via self-hosted/public OSRM + a nearest-neighbor + 2-opt heuristic
- Admin analytics: ward summaries, segregation trends, fleet status
- Rewards ledger, badges, and redemption catalog

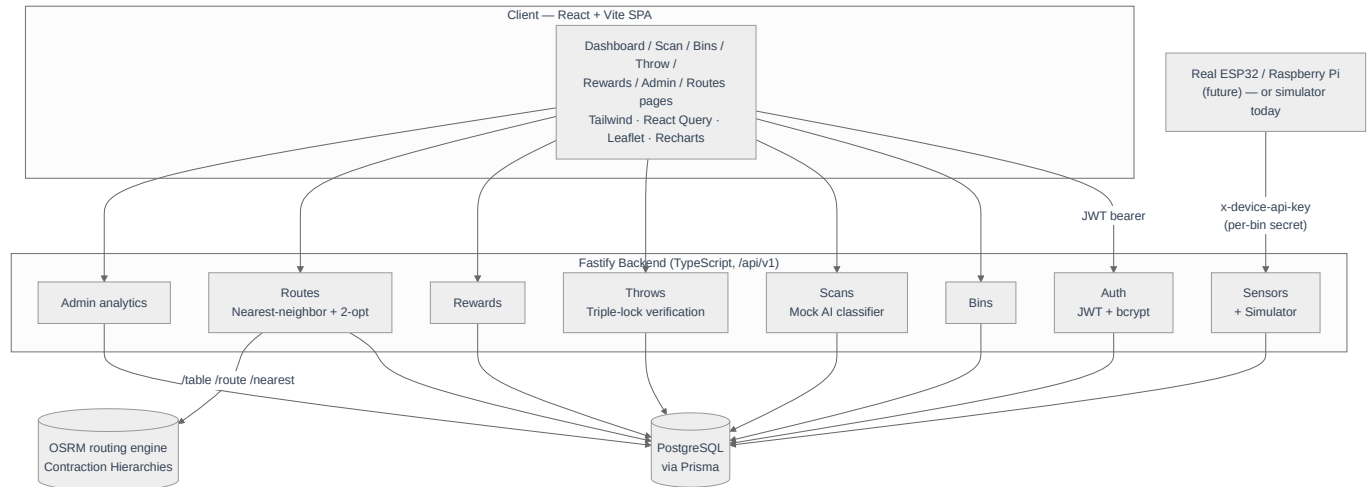
◆ Intentionally mocked / deferred

- AI waste classification: deterministic hash-based mock behind a swappable interface (no vision model / API key wired up yet)
- Raspberry Pi / ESP32 hardware: not physically connected — a software simulator posts to the exact same endpoints a real device would use
- MQTT transport: sensor ingestion uses plain authenticated HTTP POST instead, for simplicity in this prototype phase

The remainder of this report covers the implementation architecture, the database schema, the API surface, and screenshots of every major screen exercised during end-to-end verification.

2. Implementation Architecture

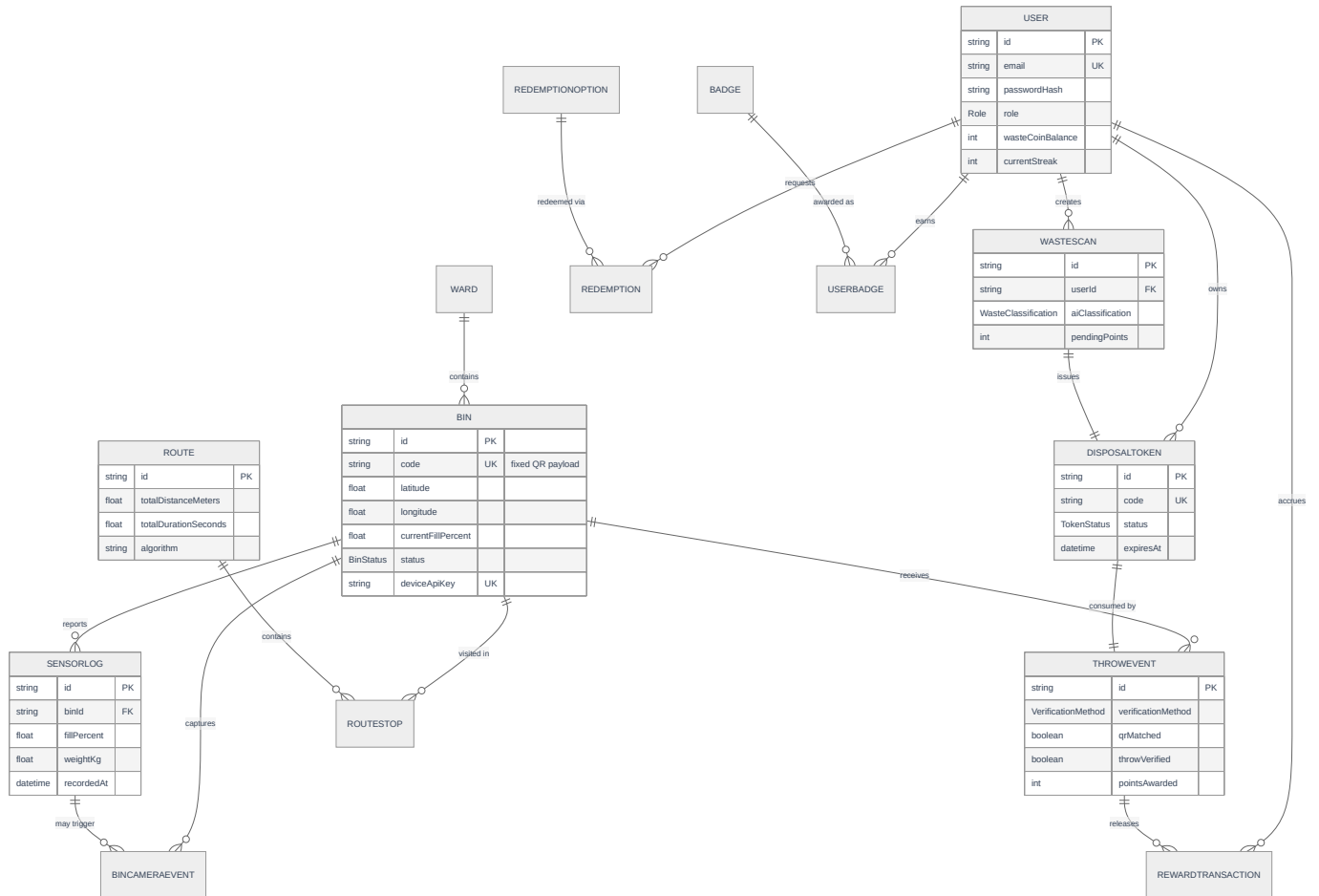
This extends the conceptual process diagram in `dustbin_arch.md` with the concrete system components built for this prototype: a Fastify backend organized as one module per domain concern, a single PostgreSQL database via Prisma, and OSRM as the real routing engine behind the route optimizer.



Hardware-ready by design: every bin has a unique `deviceApiKey`. Telemetry ingestion is authenticated via an `x-device-api-key` header — not the citizen JWT — so a real ESP32/Raspberry Pi can POST to `/api/v1/bins/:binId/telemetry` and `/api/v1/bins/:binId/camera-events` with zero backend changes. The in-process simulator used for this demo calls the identical code path.

3. Database Design

The schema mirrors the process's own state machine rather than one overloaded table, so anti-gaming failure paths (expired tokens, GPS mismatches, unverified throws) are first-class, queryable states — not just error responses.



Indexes worth noting: `SensorLog(binId, recordedAt)` for fast time-series reads, `RewardTransaction(userId, createdAt)` for the ledger view, and unique constraints on `Bin.code` / `Bin.deviceApiKey` / `DisposalToken.code` to make the QR-and-token contract enforceable at the database level.

4. API Summary

All endpoints are prefixed `/api/v1`. "JWT" means an `Authorization: Bearer` citizen or admin session token; "Device API key" means the per-bin `x-device-api-key` header used by sensors/simulator.

MODULE	METHOD	PATH	AUTH	DESCRIPTION
Auth	POST	<code>/api/v1/auth/register</code>	Public	Create a citizen account, returns JWT
Auth	POST	<code>/api/v1/auth/login</code>	Public	Verify credentials (bcrypt), returns JWT
Auth	GET	<code>/api/v1/auth/me</code>	JWT	Current user profile + balance/streak
Bins	GET	<code>/api/v1/bins</code>	JWT	List all bins with live fill state
Bins	GET	<code>/api/v1/bins/:id</code>	JWT	Single bin detail
Bins	POST	<code>/api/v1/bins</code>	JWT + ADMIN	Register a new bin
Bins	PATCH	<code>/api/v1/bins/:id</code>	JWT + ADMIN	Update bin metadata/status
Sensors	POST	<code>/api/v1/bins/:binId/telemetry</code>	Device API key	Ingest fill/weight/motion reading — same endpoint for simulator and future real hardware
Sensors	POST	<code>/api/v1/bins/:binId/camera-events</code>	Device API key	Ingest a camera capture + AI segregation score
Sensors	GET	<code>/api/v1/bins/:binId/logs</code>	JWT	Recent sensor history for charting
Sensors	GET	<code>/api/v1/bins/:binId/camera-events</code>	JWT	Recent camera events for a bin
Scans	POST	<code>/api/v1/scans</code>	JWT	Submit photo(s), run mock AI classification
Scans	POST	<code>/api/v1/scans/:id/token</code>	JWT	Issue a time-limited disposal token/QR
Scans	GET	<code>/api/v1/scans</code>	JWT	List the user's scan history
Scans	GET	<code>/api/v1/scans/:id</code>	JWT	Single scan detail
Throws	POST	<code>/api/v1/throws</code>	JWT	Triple-lock check (GPS + bin QR + token) and release of points
Rewards	GET	<code>/api/v1/rewards/transactions</code>	JWT	WasteCoin ledger
Rewards	GET	<code>/api/v1/rewards/badges</code>	JWT	Badge catalog + ownership

MODULE	METHOD	PATH	AUTH	DESCRIPTION
Rewards	GET	/api/v1/rewards/redemption-options	JWT	Available redemption catalog
Rewards	POST	/api/v1/rewards/redeem	JWT	Spend WasteCoins on a redemption option
Routes	POST	/api/v1/routes/optimize	JWT + ADMIN	Run OSRM matrix + NN/2-opt heuristic, persist a Route
Routes	GET	/api/v1/routes	JWT	List past routes
Routes	GET	/api/v1/routes/:id	JWT	Route detail with stop order + geometry
Admin	GET	/api/v1/admin/analytics/ward-summary	JWT + ADMIN	Per-ward fill % and bins-needing-collection
Admin	GET	/api/v1/admin/analytics/segregation-trends	JWT + ADMIN	Daily segregation quality trend
Admin	GET	/api/v1/admin/analytics/bins-status	JWT + ADMIN	Fleet status breakdown
Admin	POST	/api/v1/admin/simulate/tick	JWT + ADMIN	Manually advance the sensor simulator

5. Verified Demo — Citizen Journey

The full citizen-facing loop was driven end-to-end in a real browser against the running backend and a seeded PostgreSQL database, with zero console/network errors.

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Dashboard

Scan

Bins

Throw

Rewards

Aarav Sharma · 40 WC Log out

Welcome back, Aarav

Here's your WasteLoop activity at a glance.

WasteCoin balance

40

Current streak

1 days

Longest streak

1 days

Scan waste

Verify a throw

View rewards

Badges (1 earned)

First Steps

Week Warrior

Consistency Champion

Recycle Rookie

Recent activity

Full verification chain bonus

+3 WC

Points released for verified throw at bin BIN-CP-02

+15 WC

Points released for verified throw at bin BIN-KM-16

+8 WC

Full verification chain bonus

+2 WC

Full verification chain bonus

+2 WC

Points released for verified throw at bin BIN-IG-10

+10 WC

Citizen dashboard — WasteCoin balance, streaks, earned badges, recent activity ledger.

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Dashboard

Scan

Bins

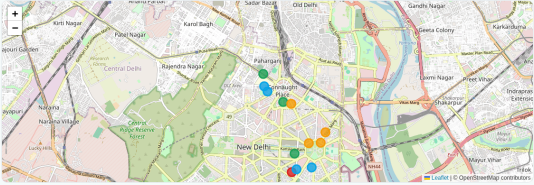
Throw

Rewards

Aarav Sharma · 40 WC Log out

Smart bins

Live fill levels reported by (simulated) IoT sensors.



Code	Label	Ward	Fill %	Status	Details
82N-CP-01	Connaught Place Bin 1	Connaught Place	0%	ACTIVE	Details
82N-CP-02	Connaught Place Bin 2	Connaught Place	69%	ACTIVE	Details
82N-CP-03	Connaught Place Bin 3	Connaught Place	69%	ACTIVE	Details
82N-CP-04	Connaught Place Bin 4	Connaught Place	10%	ACTIVE	Details
82N-CP-05	Connaught Place Bin 5	Connaught Place	80%	ACTIVE	Details
82N-IG-06	India Gate Bin 1	India Gate	77%	ACTIVE	Details
82N-IG-07	India Gate Bin 2	India Gate	68%	ACTIVE	Details
82N-IG-08	India Gate Bin 3	India Gate	78%	ACTIVE	Details

Live bin map (Leaflet) — markers colored by fill level, over the Central Delhi demo region.

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Dashboard

Scan

Bins

Throw

Rewards

Aarav Sharma · 40 WC Log out

Connaught Place Bin 1

82N-CP-01

Fill level

0%

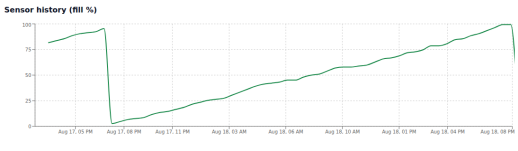
Weight

0.0 kg

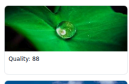
Status

ACTIVE

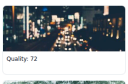
Sensor history (fill %)



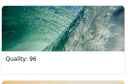
Recent camera events



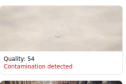
Quality 98



Quality 72



Quality 96



Quality 54
Contamination detected

Bin detail — sawtooth sensor fill-history chart plus recent camera events.

6. Verified Demo — Scan → Throw → Reward

Classify with AI

BIODEGRADABLE

89% confidence

Pending points: 8

DISPOSAL TOKEN / QR

ed51d3d8-a7af-4c68-ab64-913e26a289f5

Valid for 44m 59s

Go to nearest bin →

AI classification (mock) issues a time-limited disposal token/QR with a live countdown.

Verify disposal at a bin

Simulates the triple-lock check: GPS proximity + bin QR + valid token.

Disposal token code

ed51d3d8-a7af-4c68-ab64-913e26a289f5

Bin (scan its QR)

BIN-CP-01 — Connaught Place Bin 1

Simulate GPS at bin

Simulate wrong location

GPS latitude

28.631595

GPS longitude

77.214038

Throw verification method

Self-recorded photo

☒ Attach evidence photo (uncheck to simulate a failed capture)

Throw now

Verified! Points released 🎉

+8 points, +2 bonus WasteCoins

Triple-lock verification (GPS + bin QR + token) passes — points and bonus released instantly.

Rewards

Balance: 50 WasteCoins

Redeem

DMRC

DMRC Metro Credit

10 metro-card credits via DMRC.

120 WC

BSES

BSES Electricity Bill Discount

Rs. 50 off your next BSES electricity bill.

200 WC

BSES

BSES Electricity Bill Discount (Large)

Rs. 100 off your next BSES electricity bill.

380 WC

Transaction ledger

Date	Type	Description	Amount	Balance
8/18/2026	BONUS	Full verification chain bonus	+2	50
8/18/2026	EARN_RELEASED	Points released for verified throw at bin BIN-CP-01	+8	48
8/18/2026	BONUS	Full verification chain bonus	+3	40
8/18/2026	EARN_RELEASED	Points released for verified throw at bin BIN-CP-02	+15	37
8/18/2026	BONUS	Full verification chain bonus	+2	22
8/18/2026	EARN_RELEASED	Points released for verified throw at bin BIN-KM-16	+8	20
8/18/2026	BONUS	Full verification chain bonus	+2	12
8/18/2026	EARN_RELEASED	Points released for verified throw at bin BIN-IG-10	+10	10

Rewards page — redemption catalog (BSES/DMRC) and a fully reconciled WasteCoin transaction ledger.

7. Verified Demo — Municipal Command Center

Dashboard

Scan

Bins

Throw

Rewards

Routes

Admin

Ward Admin

0 WC

Log out

Municipal command center

Ward heatmaps, segregation trends, and fleet status.

Simulate sensor tick

Connaught Place

45.7%

5 bins · 1 need collection

India Gate

66.8%

5 bins · 3 need collection

Lodhi Road

65%

5 bins · 1 need collection

Khan Market

44.7%

4 bins · 1 need collection

Segregation quality trend (daily)

2026-08-19

Fleet status

2 17

FULL ACTIVE

Admin analytics — per-ward fill %, segregation quality trend, and fleet status.

Dashboard

Scan

Bins

Throw

Rewards

Routes

Admin

Ward Admin

0 WC

Log out

Collection route optimizer

Real road-network routing (OSRM) + nearest-neighbor/2-opt stop ordering, mirroring how OR-Tools would solve this TSP.

Optimize route (bins ≥ threshold)

Total distance

10.5 km

Total duration

14 min

Stops

6

Leaflet | © OpenStreetMap contributors

#	Bin	Fill %	Distance from prev	Duration from prev
1	India Gate Bin 1 BIN-IG-06	77%	0 m	0 min
2	Connaught Place Bin 5 BIN-CP-05	80%	2.3 km	3 min
3	India Gate Bin 3 BIN-IG-08	78%	2.7 km	3 min
4	India Gate Bin 5 BIN-IG-10	76%	490 m	1 min
5	Khan Market Bin 2 BIN-KM-17	98%	2.5 km	4 min

Route optimizer — real OSRM road-network polyline and ordered stop table, computed on demand.

8. Routing Algorithm

The route optimizer mirrors how a production system (Google Maps / OR-Tools style) would approach multi-stop vehicle routing, split into two concerns:

Real-world distances

- Candidate bins are selected by explicit ID list or by a fill-percent threshold (`ROUTE_FILL_THRESHOLD_PERCENT`)
- OSRM's `/table` endpoint returns a true road-network distance/duration matrix (Contraction Hierarchies), not straight-line estimates
- OSRM's `/route` then produces the actual driving geometry for map rendering

Visit ordering (TSP heuristic)

- Nearest-neighbor construction gives a fast initial tour
- 2-opt local search iteratively removes crossing edges to shorten the tour — the same family of technique OR-Tools' guided local search builds on
- Result is persisted as a `Route` + ordered `RouteStop` rows, with per-leg distance/duration

`OSRM_BASE_URL` is configurable — the prototype defaults to the public OSRM demo server for zero-setup development, with a one-time script (`infra/osrm/scripts/fetch_and_prepare_osm.sh`) to preprocess a small Central-Delhi OSM extract for a self-hosted instance via Docker Compose.

9. Running the Prototype

<code>npm install</code>	Install all workspace dependencies
<code>npm run db:migrate --workspace backend</code>	Apply the Prisma schema to Postgres
<code>npm run db:seed --workspace backend</code>	Seed ~19 bins, 13 users, 3 weeks of sensor history, sample throws
<code>npm run dev:backend</code>	Fastify API on :4000
<code>npm run dev:frontend</code>	Vite dev server on :5173
<code>npm run smoke --workspace backend</code>	Scripted end-to-end API verification

Demo credentials (all password `wasteloop123`): `admin@wasteloop.demo` (admin), `citizen1@wasteloop.demo` through `citizen12@wasteloop.demo`.

Nayi Disha (WasteLoop) prototype — generated from the working codebase in this repository.

